

Roll No.....

Total No. of Questions : 22]

[Total No. of Printed Pages : 8

12thARF(S/Z)JD-2026
1420-C
PHYSICS – P₃

Time : 3 Hours]

[Maximum Marks : 70

SECTION-A

(Attempt all questions. Each part carries 1 mark.)

(10×1=10)

1. Choose Most Appropriate Answer :

- (i) If an electric dipole is placed in a non-uniform electric field, it experiences :
- (a) Zero force and zero torque
 - (b) Zero force and non-zero torque
 - (c) Non-zero force and zero torque
 - (d) Non-zero force and non-zero torque

(ii) A galvanometer can be converted into an ammeter, if a low resistance called shunt is placed in

- (a) Series with galvanometer
- (b) Parallel with galvanometer
- (c) Sometimes in series and sometimes in parallel
- (d) None of them

(iii) The S.I. Unit of magnetic pole strength is

- (a) ampere/metre
- (b) ampere/metre²
- (c) ampere-metre
- (d) ampere-metre²

(iv) The S.I. Unit of capacitive reactance is

- (a) henry
- (b) ohm
- (c) farad
- (d) None of them

(v) The image formed by a compound microscope is

- (a) Real, inverted (b) Virtual, erect
(c) Virtual, inverted (d) Real, erect

(vi) What is the number of neutrons in the nucleus of ${}_{92}\text{U}^{235}$?

- (a) Zero (b) 92
(c) 143 (d) 235

(vii) Nuclear density is of the order of

- (a) 10^5 (b) 10^7
(c) 10^{10} (d) 10^{17}

(viii) Paschen series lies in which of the following region of spectrum

- (a) Ultra Violet (b) Visible
(c) Infra-Red (d) None of them

- (ix) During forward bias, the width of depletion layer
- (a) Increases
 - (b) Decreases
 - (c) First increases then decreases
 - (d) None of them
- (x) The energy gap of semi-conductors is of the order of
- (a) 1eV
 - (b) 5eV
 - (c) 10eV
 - (d) None of them

SECTION-B

(Attempt all questions. Each question carries 02 marks.)

(9×2=18)

2. Two wires of same material have lengths ℓ and 4ℓ with area of cross section $4A$ and A respectively. What is the ratio of their resistances?
3. Define drift velocity. Write the relation between current and drift velocity.

4. Define mean and effective value of alternating current. Write the relation between them.
5. Calculate the frequency of a.c at which inductive reactance of 0.7mH is 220Ω .
6. Define total internal reflection of light. Write the two conditions necessary for total internal reflection of light.
7. Define work function and threshold frequency. Give their S.I. Units.
8. Find the de Broglie wavelength of an electron accelerated through a potential difference of 100 eV . <https://www.jkboseonline.com>
9. Give two points of difference between nuclear fission and nuclear fusion.
10. Give four points of difference/similarity between p-type and n-type semi-conductor.

SECTION-C

(Attempt all questions. Each question carries 03 marks.)

(9×3=27)

11. Calculate the electric dipole moment of a dipole of length 2000°A having magnitude of each charge as 3×10^{-6} Coulomb.
12. State and explain Kirchhoff's laws.
13. Give three properties of diamagnetic, para magnetic and ferro magnetic substance. 
14. State and explain Faraday's laws of electromagnetic induction.
15. Give six properties of electromagnetic waves.
16. A ray of light suffers minimum deviation while passing through a prism of refractive index 1.5 and refraction angle of 60° . Find the angle of minimum deviation and angle of incidence.

17. What are coherent sources? Write four conditions for obtaining sustained interference of light.
18. Give three properties each of α , β and γ rays.
19. What is rectifier? How is Pn-junction diode used as a full wave rectifier?

SECTION-D

(Attempt all questions. Each question carries 05 marks.)

(3×5=15)

20. State Gauss's law of electrostatics. Find an expression for electric field due to infinitely long straight wire.

OR

Find an expression for the capacitance of a parallel plate capacitor in presence of dielectric slab between its plates.

21. State Biot Savart's law. Derive an expression for the magnetic field at the centre of circular coil carrying current.

OR

Give the principle, construction and working of moving coil galvanometer.

22. What is Lens Maker's formula? Derive an expression for Lens Maker's formula for a convex.

OR

State Huygen's Principle. Derive laws of reflection from Huygen's Principle.

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